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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
Field Sample Permit: Method details Mouse embryo collection Animal care and experiments were conducted in accordance with the Guidelines of Animal Experimentation of the National Institutes for Natural Sciences.
IACUC: All animal experiments were approved by the Animal Research Committee of the National Institutes for Natural Sciences.
Inclusion and Exclusion Criteria
not detected.
Attrition
not detected.
Sex as a biological variable
Males of R26-ZO-1EGFP (Katsunuma et al., 2016) or wild-type ICR (Japan CREA) were mated with wild-type ICR females to obtain blastocysts.
Subject Demographics
Age: not detected.
Weight: not detected.
Randomization
not detected.
Blinding
not detected.
Power Analysis
not detected.
Replication
not required.

Cell Line Authentication
not detected.
Data Information
Availability: Materials availability All unique/stable reagents generated in this study are available from the lead contact with a completed Material Transfer Agreement.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
STAR#Methods Key resources table REAGENT or RESOURCE SOURCE IDENTIFIER Antibodies anti-ZO-1 antibody Thermo Fisher Scientific cat. # 33-9100; RRID:AB_2533147 goat anti-mouse IgG antibody, Alexa Fluor 488 Thermo Fisher Scientific cat. # A28175; RRID:AB_2536161 anti-Claudin-1 antibody Thermo Fisher Scientific cat. # 51-9000;RRID: AB_2533916 goat anti-rabbit IgG antibody, Alexa Fluor 555 Thermo Fisher Scientific cat. # A27039; RRID:AB_ 2536100 Chemicals, peptides, and recombinant proteins Blocking One Nacalai Tesque cat.#	anti-ZO-1		
			(Thermo Fisher Scientific Cat# 33-9100, RRID:AB_2533147)(link)
	anti-mouse IgG		
	anti-mouse IgG Alexa Fluor	Thermo Fisher Scientific	(Thermo Fisher Scientific Cat# A28175, RRID:AB_2536161)(link)
	anti-Claudin-1	Thermo Fisher Scientific	(Thermo Fisher Scientific Cat# 51-9000, RRID:AB_2533916)(link)
	anti-rabbit IgG Alexa Fluor	Thermo Fisher Scientific	(Thermo Fisher Scientific Cat# A27039, RRID:AB_2536100)(link)
A22283; RRID:AB_2632953 Hoechst 33342 Thermo Fisher Scientific cat.#			(Thermo Fisher Scientific Cat# A-22283, RRID:AB_2632953)(link)
The following procedure was the same as described above, except that we used 50-times diluted anti-claudin-1 antibodies (cat. # 51-9000, Thermo Fisher Scientific) and 200-times diluted goat anti-rabbit IgG antibody Alexa Fluor 555 (cat. # A27039, Thermo Fisher Scientific).	anti-claudin-1		
	anti-rabbit IgG		
Experimental Models: Cell Lines			
MDCK cells were cultured in Dulbecco's modified Eagle medium (DMEM) containing 10% FBS.	MDCK		Suggestion: RRID:CVCL_0422(link)
Software and Algorithms			
# 087-00432 Experimental models: organisms/strains R26-ZO-1EGFP RIKEN accession # CDB026K wild type ICR mice Japan CREA N/A Experimental models: cell lines MDCK I cells Fujiwara et al. (2016) N/A A6 cells Mimori-Kiyosue et al. (2007) N/A Recombinant DNA human ZO-1 Konno et al. (2008) N/A GFP-ZO-1/pCS2 This paper N/A GFP-ZO-1ΔABD/pCS2 (Δ1151-1371) This paper N/A GFP-ZO-1ΔABD-ΔIDR123/pCS2 (Δ114-118, Δ275-415, Δ588-627, Δ1151-1371) This paper N/A GFP-	Fiji		
	https://fiji.sc		
	Python		Suggestion: python-bx (RRID:SCR_024202)(link)
	https:// www.python.org		

ZO-1ΔIDR4/pCS2 (Δ858-1580) This paper N/A GFP-ZO-1-PSG/pCS2 (424-794) This paper N/A GFP-ZO-1ΔABDΔPSG/pCS2 (Δ424--794, Δ1151-1371) This paper N/A Lifeact-RFP Iioka et al. (2007) N/A RFP-MoesinABD Iioka et al. (2004) N/A mCherry-Utrophin Kim and Davidson (2013) N/A Software and algorithms Fiji NIH https://fiji.sc Python 3.9 Python Core Team https://www.python.org Pandas ver.			
0.11.1 https://seaborn.pydata.org Matplotlib ver.	Matplotlib		
Measurement of fluorescent intensities and particle analyses Fiji/Image J was used to measure fluorescent intensities.	Fiji/Image J		
Then we conducted a line plot analysis in Image J software.	Image J		Suggestion: ImageJ, RRID:SCR_003070(link)
Laser ablation Laser ablation was conducted using an Olympus IX 81 inverted microscope (20 × /0.70 NA dry objective lens), equipped with a spinning-disc confocal unit Yokogawa CSUX-1 and iXon3 897 EM-CCD camera (Andor), controlled with Andor IQ2 software.	Andor IQ2		
We performed two-tailed Student's t-test, Mann-Whitney U-test, and Tukey-Kramer test for the statistical analyses, as indicated in the figure legends, using Python scipy and statsmodels packages.	scipy		

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
We performed two-tailed Student's t-test, Mann-Whitney U-test, and Tukey-Kramer test for the statistical analyses, as indicated in the figure legends, using Python scipy and statsmodels packages.	Student's t-test
	Mann-Whitney U-test
	Tukey-Kramer test

SciScore is an automated tool that is designed to assist expert reviewers by finding and presenting formulaic information scattered throughout a paper in a standard, easy to digest format. ***SciScore is not a substitute for expert review.*** SciScore also checks for the presence and correctness of several unique identifiers, including RRIDs (research resource identifiers) in the manuscript, detects sentences that appear to be missing RRIDs, and can even suggest RRIDs under certain circumstances. **All RRID suggestions should be verified;** only the author can know whether the suggestions are correct.

For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq